MIL-S-36019A
7 September 1978
SUPERSEDING
MIL-S-36019
25 May 1959

MILITARY SPECIFICATION

STRAP, WEBBING, PATIENT SECURING

This specification is approved for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1.2 Scope This specification covers one type of patient securing strap (see 6.1)

2 APPLICABLE DOCUMENTS

2.1 Issues of documents The following documents of the issue in effect or date of invitation for bid or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-285	- Thread, Polyester
QQ-A-591	- Aluminum Alloy, Die Castings
QQ-N-290	- Nickel Plating (Electrodeposited)
QQ-W-470	- Wire, Steel, Carbon, Spring, Music
TT-E-527	- Enamel, Alkyd, Lustreless
PPP-B-636	- Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-MTT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426, appearing at the end of this document or by letter

FSC 6530

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MILITARY

MIL-E-495 - Finish, Chemical, Black, For Copper Alloys
MIL-C-496 - Clip, End, Strap
MIL-W-530 - Webbing, Textile, Cotton, General Purpose, Natural or Ir Colors
MIL-S-6050 - Steel, CHROME-NICKEL-MOLYBDENUM (8630) Bars and Reforging Stock (Aircraft Quality)
MIL-S-18729 - Steel Plate, Sheet, and Strip Alloy, 4130, Aircraft Quality

STANDARDS

FEDERAL

FED-STD-66 - Steel, Chemical Composition and Hardenability
FED-STD-595 - Colors
FED-STD-751 - Stitches, Seams and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS

10125 - Strap, Webbing, Patient Securing

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3 REQUIREMENTS

3.1 First article When specified in the procurement document the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.1.2.1 and 6.2).

3.2 Materials

3.2.1 Buckle

3.2.1.1 Buckle frame Buckle frame shall be made from 4130 steel conforming to MIL-S-18729 or carbon steel AISI 1008 to 1040 as referenced in FED-STD-66; having a minimum (min) tensile strength of 80,000 psi

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3 2 1 2 Buckle pin The buckle pin shall be made from carbon steel, AISI 1030 to 1040, as referenced in FED-STD-66, tempered to a hardness of 30 to 40 on the Rockwell C scale, or 8630 steel hardened and tempered in accordance with MIL-S-6050

3 2 1 3 Latch Latch shall be die cast aluminum alloy conforming to QQ-A-591

3 2 1 4 Springs The wire spring and loading spring shall be 0 036 inch diameter music wire conforming to QQ-W-470, having a hardness of 47 to 52 on the Rockwell C scale

3 2 2 webbing Webbing shall be 2 inches wide conforming to type III, class 4 of MIL-W-530 The webbing shall be vat-dyed to approximately olive drab No 7

3 2 3 Thread Thread used for stitching shall be soft, black thread conforming to Type I or II, Class 1, size FF, of V-T-285

3 2 4 Plastic tip When a plastic web tip is used it shall be made from a black, durable plastic, resistant to breakage

3 2 5 Metal tip When a metal web tip is used, the tip shall be type II, class 1 2 inch size in accordance with MIL-C-496 Finish shall be a black chemical finish in accordance with MIL-F-495

3 3 Construction The patient securing strap shall be as specified herein and in accordance with DPSC Drawing 10125 The complete strap shall be tested as specified in 4 5 6, the webbing shall not slip in the buckle and the latch of the buckle shall tighter under the pull of the webbing The buckle shall show no distortion, nor shall the buckle cut the webbing during this test The buckle shall not work open when operated by hand as specified in 4 5 6

3 3 1 Buckle Buckle frame shall be formed from steel sheet with cutout for webbing loop Locking device shall be an aluminum alloy, spring-actior lever with serrated gripping surface, pivoting on a steel pin, and with lever release button as shown on the drawing The load required to deflect the flat type of loading spring to 0 080 inch shall be 2 pounds ± 1 ounce and to 0 050 inch shall be 6 pounds ± 2 ounces, when tested as specified in 4 5 4 Assembly shall be such that all buckles will pass the opening and closing test outlined in 4 5 5

3 3 2 Strap All raw ends of the webbing shall be treated to prevent ravelling The strap shall be cut, and assembled to comply with dimensional requirements of DPSC Drawing 10125

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3 3 2 1 Government furnished webbing When specified in the contract (see 6.2) a sufficient quantity of webbing for use in the manufacture of the strap will be furnished by the Government The quantity and terms of this Government furnished material (GFM) will be specified in the contract When GFM is specified, the contractor is required to conduct visual examination of the material to insure it is free of defects as specified in MIL-W-530

3 3 3 Stitching All stitching shall be machine stitching conforming to stitch type 301 of FED-STD-751 Stitching shall have 7-9 stitches per inch All breaks in stitching and stitching ends shall be backstitched not less than 1 inch

3 4 Finish Steel parts shall be nickel plated in accordance with class 1, Grade E of of QQ-N-290 Aluminum parts shall be finished by anodizing in accordance with type 2, class 1 of MIL-A-8625 Finished parts shall be thoroughly cleaned after plating The assembled buckle shall be coated with baked or enamel in accordance with class B of TT-E-527 The color of the enamel shall be 34087 in acordance with FED-STD-595 The retaining ring on the buckle need not be painted

3 5 Identification marking The manufacturers name or registered trademark shall be stenciled on the undersurface of the webbing of the strap with black, waterproof, stencil paint, or with black, waterproof ink The buckle shall be permanently and legibly marked with the same information

3 6 Workmanship Workmanship shall be first class throughout Item shall be free from defects which detract from its appearance or impair its serviceability All burrs and sharp edges shall be removed from the item

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

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4 1 1 Records Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate

4 1 2 Inspection Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties of the item)

4 1 2 1 Classification of inspection The inspection requirements specified herein are classified as follows

a First article inspection
b Quality conformance inspection

4 1 2 1 1 First article inspection First article inspection shall be performed on the strap when a first article sample is required (see 3 1) This inspection shall include examination of 4 3 and tests of 4 4

4 1 2 1 2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4 3 and tests of 4 4

4 1 3 Certificates of quality Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements

4 2 Inspection lot

4 2 1 Raw materials An inspection lot for raw material inspection shall be all material produced under the same conditions and controlled to the degree that uniformity of product within the designated lot can be assured Lot identification of the raw material by the manufacturer is one method of control, as cited above, which is considered acceptable

4 2 2 End product An inspection lot for inspection of end product shall consist of all straps of the same type produced under essentially the same conditions and offered for inspection at one time

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4 3 Sampling

4 3 1 For examination Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I The unit of product for sampling purposes shall be a strap

TABLE I Sampling for examination

	Inspection level	AQL (defect per hundred units)
For visual examination		
Major defects	II	2 5
Total defects (Major and Minor combined)	II	6 0
For dimensional defects	S-1	4 0

4 3 2 Tests Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II, for an AQL of 1 0

TABLE II Sampling for tests

Component (Unit of product)	Characteristic	Requirement	Test procedure	Inspection level
Webbing	-	3 2 2	4 5 2	-
Thread	-	3 2 3	4 5 3	-
Loading Spring	Load deflection	3 3 1	4 5 4	S-2
Buckle	Manipulation	3 3 1	4 5 5	S-2
End Item	Performance	3 3	4 5 6	S-2

4 3 3 Raw materials A sufficient quantity of each metal used in the manufacture of the components shall be selected for chemical analysis

4 4 Examination The strap shall be examined to determine compliance with all requirements contained in this specification

4 4 1 Classification of defects Examination shall be conducted in accordance with the classification of defects listed in table III Examination shall not be restricted to the classified possible defects listed below

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TABLE III Classification of defects

Major	
101	Strap not complete with all component parts
102	Metal or plastic parts not free from burrs, sharp edges, etc
103	Ends of stitch lines not backstitched
104	Strap not free from holes, tears, or cuts which may weaken the strap
105	Tip missing or not properly attached to webbing
106	Buckle fails to function
107	Tip not firmly secured to webbing
108	Stitching type not as specified
109	Breaks in stitching not backstitched
Minor	
201	Number of stitches per inch not as specified
202	Identification marking not present or legible
203	Plastic tip not black
204	Finish scratched, chipped, etc
205	Free end of strap not properly finished to prevent raveling
206	Color of buckle and webbing not as specified

4.4.2 Dimensional examination The strap shall be examined for defects in dimensions Any dimension not within the tolerances specified herein shall be classified as a defect

4.5 Tests Tests shall be conducted to determine compliance with specification requirements

4.5.1 Chemical composition Four ounces from each lot material shall be tested in accordance with FED-TEST METHOD STD-151

4.5.2 Webbing The webbing shall be tested as specified in MIL-W-530

4.5.3 Thread The thread shall be tested in accordance with V-T-285

4.5.4 Load deflection test The load shall be uniformly applied perpendicularly to the flat surface of the spring The load required to deflect the spring the specified distance shall be reported

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4.5.5 Manipulation of buckle. With the end of the strap not inserted in the buckle, the following tests shall be made:

a. With the curved surface of the latch resting against the buckle, the latch shall be turned into the open position for insertion of the strap. No sidewise pressure shall be needed to force the latch pin into the mating hole in the buckle frame.

b. With the latch in the open position (a) the hexagonal head of the buckle pin shall be pressed. The latch shall go back into the original position (curved surfaces resting against the buckle) without further assistance.

4.5.6 Performance test. The strap shall be inserted through the buckle, the latch closed, and the buckle subjected to a 500 pound pull to determine strength and fit of webbing and buckle. After the pull test, the strap shall be taken with one hand on each side of the buckle, and the webbing moved toward and away from the buckle to determine whether the buckle will work open. The hand shall be placed on the webbing with the inside of the hand 4 inches from the buckle. Any failure shall be cause for rejection. Straps subjected to this test shall be discarded.

4.6 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Unit of issue. One strap constitutes a unit of issue.

5.1.2 Level A. Each strap shall be clean, dry, tightly wound and secured with cord or by other suitable means. Each strap shall be further packaged by being wrapped in kraft paper and the wrapped strap secured with kraft gummed tape. As an alternate, each strap may be shrink wrapped with a plastic film.

5.1.3 Level C. Each strap shall be clean, dry and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

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5 2.1 Level A. Seventy-two unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant Closure, strapping and waterproofing shall be as specified in the appendix of the box specification

5 2 2 Level B Seventy-two unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic Closure shall be as specified in the appendix of the box specification

5 2 3 Level C The packaged straps shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity These packs shall conform to the carrier's rules and regulations

5 3 Marking Unit packages and shipping containers shall be marked as specified in MIL-STD-129

6 NOTES

6 1 Intended use The strap covered by this specification is intended for securing patients to litters. NSNs 6530-00-783-7205 and 6530-00-783-7905

6 2 Ordering data Purchasers should select the preferred options permitted herein and include the following information in procurement documents

a Title, number and date of this specification
b Selection of applicable levels of packaging and packing (see 5 1 and 5 2)
c When a first article is required (see 3 1, 4 1 2 1 and 6 3)
d When government furnished material is required (see 3 3 2 1 and 6 6)

6 3 First article When a first article is required, it shall be tested and approved under the appropriate provisions of the Defense Acquisition Regulation The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article

6 4 This specification covers the following item appearing in the Federal Supply Catalog

National Stock Number	Item Identification
6530-00-784-4205	STRAP, WEBBING, PATIENT SECURING

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6.5 Changes from previous issue Asterisks are not used in this revision to identify changes with respect to previous issue, due to the extensiveness of the changes

6.6 Government furnished materiel When GFM is specified in the contract, the contract shall specify the quantity and terms of the GFM to be delivered

6.7 Availability of color chips Color chips are available by referencing the color number and FED-STD-595, from

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5801 Tabor Avenue
Philadelphia, PA 19120

Custodians
Army - MD
Navy - MS
Air Force - 03

Review activity
DOD - MB

Preparing activity
DLA - DM

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DOCUMENT IDENTIFIER AND TITLE
MIL-S-36019A STRAP, WEBBING, PATIENT SECURING

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
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